MATHEMATICS

Analysis and approaches

Higher Level | Set B | Paper 2

Mock examination · 2025 series

Time allowed: 2 hours Maximum mark: 110

Candidate name: _____

Candidate number: _____

Date: _____

Instructions

- A graphic display calculator is required.
- This paper has Section A and Section B. Both sections are compulsory.
- Answer every question. Show all working and give sufficient mathematical reasoning.
- A clean Mathematics: analysis and approaches formula booklet may be used.
- Give exact answers where appropriate. Otherwise give numerical answers to three significant figures unless instructed differently.
- Diagrams are not necessarily drawn to scale. Use the stated measurements and conditions.
- The number in square brackets shows the maximum mark for each part.

SECTION A

Question 1 [Maximum mark: 6]

A fair coin is tossed twelve times independently. Let X be the number of tails.

(a) Find the probability that the numbers of heads and tails are equal. [3]

(b) Find the probability that there are more tails than heads. [3]

SECTION A

Question 2 [Maximum mark: 6]

The paired data are shown below. The unknown value a is a whole number. A rounded equation of the least-squares regression line of y on x is $y = 2.398x + 101.446$.

x	12	15	17	28	a
y	127	142	144	171	151

(a) Find $\bar{y}$. [1]

(b)(i)) Use the regression line and the fact that $(\bar{x}, \bar{y})$ lies on it to estimate $\bar{x}$, giving your result to the nearest whole number. [2]

(b)(ii)) Hence find a. [2]

(c) Calculate the product-moment correlation coefficient. [1]

SECTION A

Question 3 [Maximum mark: 7]

Initially consider $f(x) = 2x^2 + 20x + 54$ for all real x.

(a) State the range of f. [1]

(b) The domain is now restricted to $x \leq k$. Explain why a restriction is needed for an inverse function, and find the largest possible value of k. [2]

(c) Using this value of k, find $f^{-1}(x)$, stating its domain. [4]

SECTION A

Question 4 [Maximum mark: 9]

All trigonometric inverse functions in this question use radians.

(a) Using the substitution $u = x^2$, find $\int \frac{x}{1+x^4}\,dx$. [4]

(b) Evaluate $\int_0^1 \frac{x}{1+x^4}\,dx$. [1]

(c) Write $\int_0^1 \frac{2x+k}{1+x^4}\,dx$ in the form $A + Bk$, and find numerical values of A and B to four significant figures. [4]

SECTION A

Question 5 [Maximum mark: 6]

Assume $|x| < 2$.

(a) Find the first three terms, in ascending powers of x, of $(4 + 2x)^{-1/2}$. [4]

(b) Hence obtain the first three non-zero terms of $\frac{64x}{\sqrt{4 + 2x}}$. [2]

SECTION A

Question 6 [Maximum mark: 6]

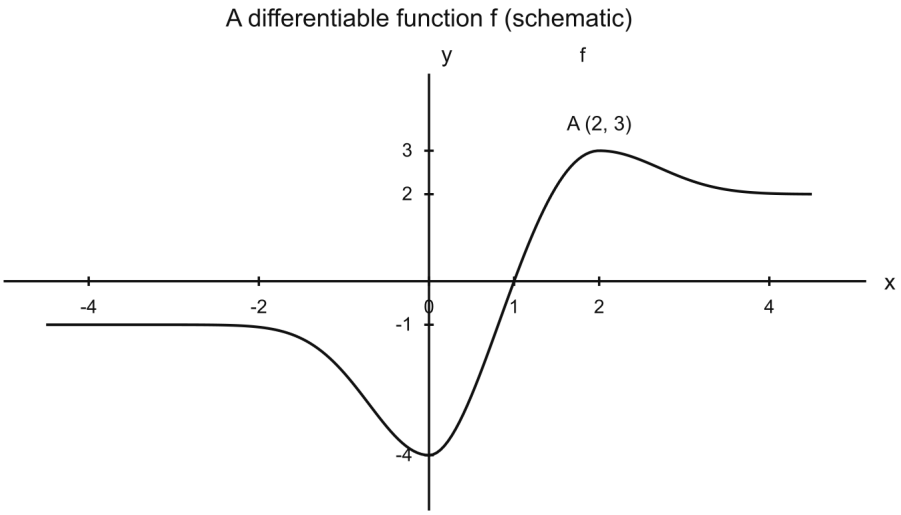
Angles are measured in radians. Consider the equation $\cos(\pi x/10) = 0$ for $0 \leq x \leq 1000$.

Find the sum of all solutions in this interval. [6]

SECTION A

Question 7 [Maximum mark: 6]

The diagram shows a differentiable function f. Point $A(2, 3)$ is a stationary maximum. Define $g(x) = x^2 + 1$ and $h(x) = f(x)/g(x)$.



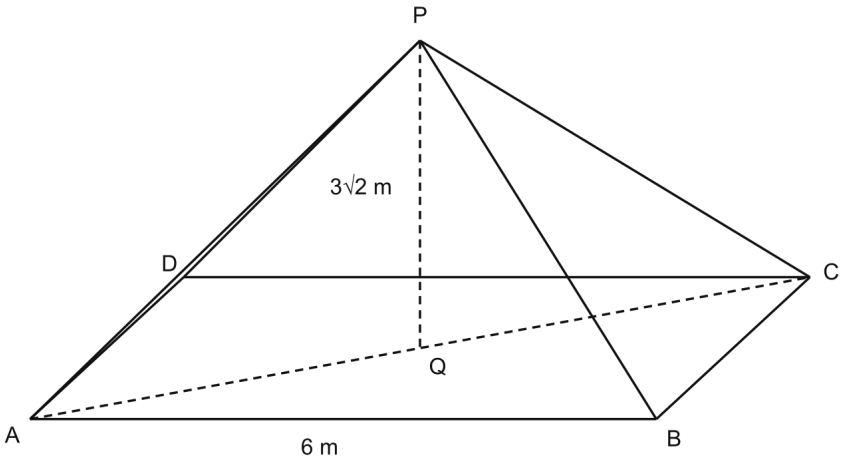
Find an exact equation of the normal to the graph of h at $x = 2$. [6]

SECTION A

Question 8 [Maximum mark: 8]

$ABCD$ is a square of side 6 metres and Q is its centre. Point P is vertically above Q, with $PQ = 3\sqrt{2}$ metres. Two ants start at P simultaneously: one moves along PA at $1.5\,\text{cm}\,\text{s}^{-1}$, and the other along PB at $2\,\text{cm}\,\text{s}^{-1}$. Their speeds remain constant.

Square-based pyramid; lengths shown in metres



(a) Show that triangle PAB is equilateral. [3]

(b) Find the rate at which the distance between the ants is increasing after 40 seconds. Work in centimetres and seconds. [5]

SECTION B

Question 9 [Maximum mark: 18]

Three planes are defined by $\Pi_1 : x - y + 6z = 1$, $\Pi_2 : 2x + y = 5$, and $\Pi_3 : 3x + y + az = b$, where a, b are real constants. Let $L = \Pi_1 \cap \Pi_2$.

(a) Find the acute angle between Π_1 and Π_2, in degrees. [3]

(b) Find the acute angle between the positive x-axis and Π_1, in degrees. [3]

(c) Find a vector equation of L. [2]

(d) When $a = 4$ and $b = 14$, find the common point of all three planes. [1]

(e)(i)) Use row operations or equivalent algebra to find a and b for which the three planes share infinitely many points. [4]

(e)(ii)) For these values, state the set of common points. [1]

(f)(i)) Find the conditions on a, b for which there is no common point. [2]

(f)(ii)) Under the conditions in part(f)(i), justify that L is parallel to Π_3. [2]

SECTION B

Question 10 [Maximum mark: 20]

The mass R of a standard packet, in grams, is normally distributed with mean 500 and standard deviation 18. Packet masses are independent.

(a) Find $P(R < 480)$. [1]

(b) Find $P(R < 500 \mid R > 480)$. [2]

(c) Find c such that $P(R > c) = 0.15$. [2]

(d) Five standard packets are selected independently. Find the probability that exactly one has mass below 480 grams. [3]

(e)(i)) The mass C of a premium packet is also normal. Its lower quartile is 724 grams and its interquartile range is 52 grams. Find its mean. [2]

(e)(ii)) Find the standard deviation of C. [3]

(f)(i)) A further normal random variable X has $E(X) = 400$ and $\mathrm{Var}(X) = 25$. Find $E(X^2)$. [2]

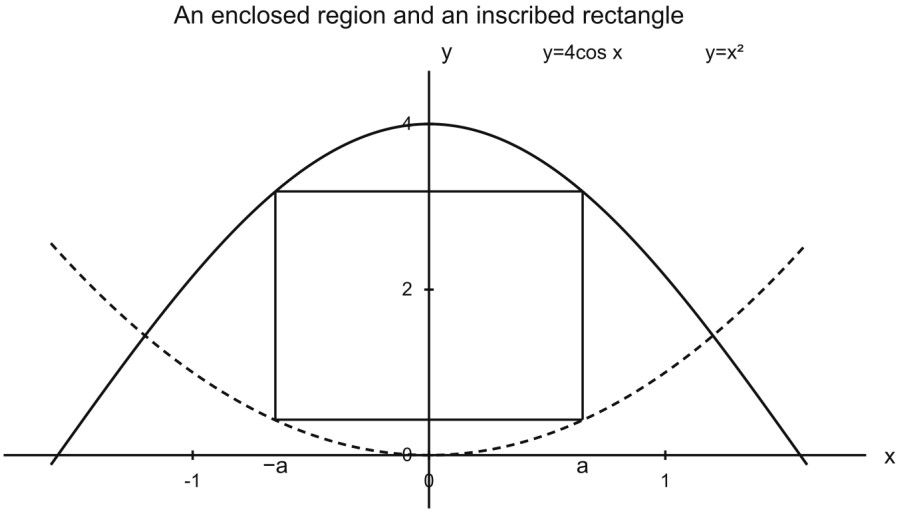
(f)(ii)) Let $Y = 3X + 2$. Find $E(Y)$ and $\mathrm{Var}(Y)$. [3]

(f)(iii)) Let $Z = (X - 400)/5$. State $E(Z)$ and $\mathrm{Var}(Z)$. [2]

SECTION B

Question 11 [Maximum mark: 18]

The curves $y = 4\cos x$ and $y = x^2$ enclose a finite region R around the y-axis. Angles are in radians. A rectangle inside R has vertical sides at $x = -a$ and $x = a$, with its upper vertices on $y = 4\cos x$ and lower vertices on $y = x^2$. The diagram is schematic.



(a) Find the coordinates of the two intersections of the curves. [2]

(b) Find the area of R. [3]

(c) Find the volume generated when R is rotated through 2π about the x-axis. [3]

(d) Find the volume generated when R is rotated through 2π about the y-axis. [4]

(e) Show that the area of the rectangle is $A(a) = 8a\cos a - 2a^3$, and state the physically relevant interval for a. [2]

(f)(i)) Show that the value of a for the maximum area satisfies $\frac{\cos a - a \sin a}{a^2} = \frac{3}{4}$. [2]

(f)(ii)) Find the maximum area of the rectangle. [2]

End of question paper

MATHEMATICS

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Worked solution book

This section contains the worked solutions and marking points. Use it after completing the question paper. Award method, accuracy and reasoning marks separately, and accept mathematically equivalent valid methods.

Mark your written working against the solution book before entering self-assessed marks in the portal. A correct final answer does not always earn every method or proof mark.

Section A

1. [Maximum mark: 6]

(a) Binomial distribution with $n = 12$, $n = \frac{1}{2}$ **A1**

$\mathrm{P}(X = 6) = 0.22558... \cong 0.226$ **M1 A1**

[3 marks]

(b) **METHOD A**

$\mathrm{P}(X \geq 7) \cong 0.387$ **M2 A1**

METHOD B

Probability of different scores: $1 - 0.22558... \cong 0.77442...$ **M1**

Probability of obtaining **more** tails than heads (due to symmetry) **M1**

$\frac{0.77442...}{2} \cong 0.387$ **A1**

[3 marks]

(a) $\overline{y} = 147$. **A1**

[1 mark]

(b) (i) the point $(\overline{x}, \overline{y})$ is on the regression line, hence

$\overline{y} = 2.398\overline{x} + 101.446 \Leftrightarrow$ **M1**

$\Leftrightarrow 147 = 2.398\overline{x} + 101.446$

$\Leftrightarrow \overline{x} = 19$ **A1**

(ii) **EITHER** by trial and error

OR by $\overline{x} = 19 = \frac{12 + 15 + 17 + a + 28}{5} \Leftrightarrow a = 23$. **M1 A1**

[4 marks]

(c) Find the correlation coefficient $r = 0.964$. **A1**

[1 mark]

$f(x) = 2x^2 + 20x + 54$, $x \le k$.

(a) Vertex at $(-5, 4)$ (by GDC graph)

Range: $y \ge 4$ **A1**

[1 mark]

(b) The curve does not satisfy the horizontal line test. **R1**

$k = -5$ **A1**

[2 marks]

(c) Vertex form: $f(x) = 2(x+5)^2 + 4 = y$ **M1**

$(x+5)^2 = \frac{y-4}{2}$ **A1**

$x + 5 = -\sqrt{\frac{y-4}{2}}$ **A1**

$x = -5 - \sqrt{\frac{y-4}{2}}$

$f^{-1}(x) = -5 - \sqrt{\frac{x-4}{2}}$ **A1**

[4 marks]

(a) $u = x^2 \Rightarrow \frac{du}{dx} = 2x \Rightarrow dx = \frac{du}{2x}$ **M1**

$\int \frac{x}{x^4+1} dx = \int \frac{x}{u^2+1}\frac{du}{2x} = \frac{1}{2}\int \frac{1}{u^2+1} du = \frac{1}{2}\arctan u + c$ **A1 A1**

$= \frac{1}{2}\arctan x^2 + c$ **A1**

[4 marks]

(b) $\int_0^1 \frac{x}{x^4+1} dx = \left[\frac{1}{2}\arctan x^2\right]_0^1 = \frac{1}{2}\arctan 1 - \frac{1}{2}\arctan 0\,.$

$\frac{\pi}{8}$ OR $0.39269... \cong 0.393$, directly by GDC. **A1**

[1 mark]

(c) $\int_0^1 \frac{2x+k}{x^4+1} dx = \int_0^1 \frac{2x}{x^4+1} dx + k\int_0^1 \frac{1}{x^4+1} dx = A + Bk\,,$ **M1**

$A = \int_0^1 \frac{2x}{x^4+1} dx\,, \quad B = \int_0^1 \frac{1}{x^4+1} dx$ **M1**

$A \cong 0.7854$ **A1**

$B \cong 0.8670$ **A1**

[4 marks]

(a) $(4+2x)^{-\frac{1}{2}} = 4^{-\frac{1}{2}}\left(1+\left(-\frac{1}{2}\right)\left(\frac{x}{2}\right)+\frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{2}\left(\frac{x}{2}\right)^2+\cdots\right)$ **M1 A1 A1**

$= \frac{1}{2}\left(1-\frac{1}{4}x+\frac{3}{32}x^2+\cdots\right) = \frac{1}{2}-\frac{1}{8}x+\frac{3}{64}x^2+\cdots$ **A1**

[4 marks]

(b) $f(x) = \frac{64x}{\sqrt{2x+4}} = 64x\left(\frac{1}{2}-\frac{1}{8}x+\frac{3}{64}x^2+\cdots\right)$ **M1**

$= 32x-8x^2+3x^3+\cdots$ **A1**

[2 marks]

$\cos\frac{\pi x}{10}=0$

Solving **EITHER** analytically **OR** by GDC, the solutions are

5, 15, 25, …, 995 (**OR** $x=5+10k$, with k from 0 to 99) **A2 A1**

Then

METHOD A (Arithmetic sequence) **M1**

Arithmetic sequence, $S_{100}=50,000$ **A2**

METHOD B (using Sigma notation) **M1**

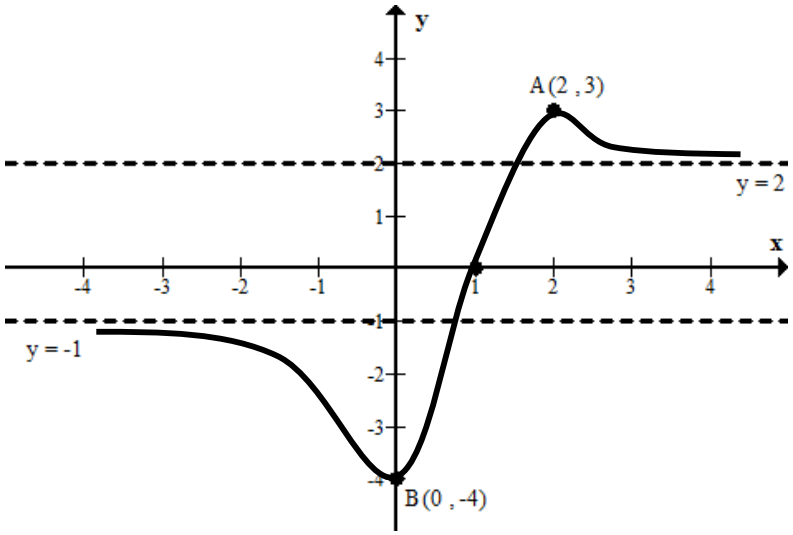
$\sum_{x=0}^{99}(5+10x)=50,000$ **A2**

METHOD C **M1**

Using Recursion: $a_n=5+10n$, sum $=50,000$ **A2**

[6 marks]

$f(x)$:



$g(x) = x^2 + 1$

For $x = 2$, $y = \frac{f(2)}{g(2)} = \frac{3}{5}$ **A1**

$\frac{dy}{dx} = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}$ **M1**

$g'(x) = 2x$ **A1**

$\left.\frac{dy}{dx}\right|_{x=2} = \frac{f'(2)g(2) - f(2)g'(2)}{g(2)^2}$

$= \frac{0 - 3 \times 4}{5^2} = -\frac{12}{25}$ **A1**

$m_N = \frac{25}{12}$ **M1**

$y - \frac{3}{5} = \frac{25}{12}(x - 2)$ **A1**

[6 marks]

(a) $AC = \sqrt{6^2 + 6^2} = 6\sqrt{2}$ **M1 A1**

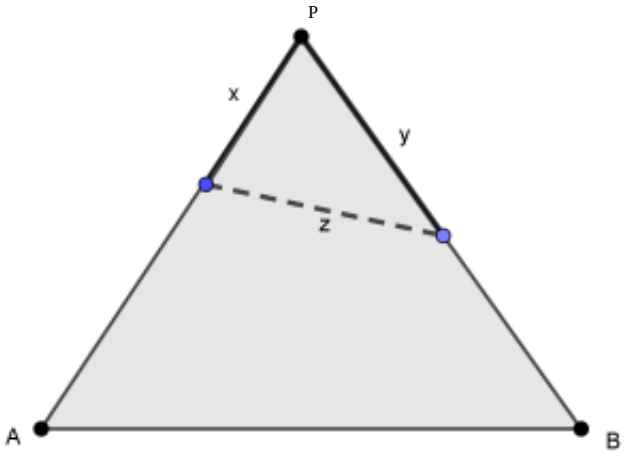
$AQ = 3\sqrt{2}$

$PA = PB = \sqrt{18+18} = 6$ **A1**

Hence the triangle PAB is equilateral.

[3 marks]

(b) Let x and y be the distances run by the two ants towards A and B respectively.



Let z be the distance between the ants.

$\frac{dx}{dt} = 1.5$, $\frac{dy}{dt} = 2$ and look for $\frac{dz}{dt} = ?$

Using Cosine rule: $z^2 = x^2 + y^2 - 2xy\cos 60° \Rightarrow z^2 = x^2 + y^2 - xy$ **A1**

Differentiate in terms of t (time).

$$2z\frac{dz}{dt} = 2x\frac{dx}{dt} + 2y\frac{dy}{dt} - x\frac{dy}{dt} - y\frac{dx}{dt}$$ **M1 A1**

After 40 seconds: $x = 60$, $y = 80$, thus $z^2 = 60^2 + 80^2 - 60\times 80 \Rightarrow z = 20\sqrt{13}$ **A1**

$$2\times 20\sqrt{13}\frac{dz}{dt} = 2(60)(1.5) + 2(80)(2) - 60(2) - (80)(1.5)$$

$$\frac{dz}{dt} = \frac{\sqrt{13}}{2} \ \mathrm{cm\,s^{-1}}$$ **A1**

[5 marks]

9. [Maximum mark: 18]

(a) $\cos\theta = \dfrac{\begin{pmatrix}1\\-1\\6\end{pmatrix}\bullet\begin{pmatrix}2\\1\\0\end{pmatrix}}{\sqrt{1+1+36}\times\sqrt{4+1}} = \dfrac{1}{\sqrt{38}\times\sqrt{5}} \Rightarrow \theta \cong 85.8°$ **M1 A1 A1**

[3 marks]

(b) $\sin\theta = \dfrac{\begin{pmatrix}1\\-1\\6\end{pmatrix}\bullet\begin{pmatrix}1\\0\\0\end{pmatrix}}{\sqrt{1+1+36}\times\sqrt{1}} = \dfrac{1}{\sqrt{38}} \Rightarrow \theta \cong 9.34°.$ **M1 A1 A1**

[3 marks]

(c) Directly from GDC (simultaneous): $r = \begin{pmatrix}2\\1\\0\end{pmatrix} + \lambda\begin{pmatrix}-2\\4\\1\end{pmatrix}$ **A1 A1**

[2 marks]

(d) Directly from GDC (simultaneous): $(x, y, z) = (-5, 15, 3.5)$ **A1**

[1 mark]

(e) (i) $\begin{pmatrix}1&-1&6&1\\2&1&0&5\\3&1&a&b\end{pmatrix} \sim \begin{pmatrix}1&-1&6&1\\0&3&-12&3\\0&4&a-18&b-3\end{pmatrix} \begin{matrix}\\L_2-2L_1\\L_3-3L_1\end{matrix}$ **M1 A1**

$\sim \begin{pmatrix}1&-1&6&1\\0&1&-4&1\\0&4&a-18&b-3\end{pmatrix} L_2 \div 3 \sim \begin{pmatrix}1&-1&6&1\\0&1&-4&1\\0&0&a-2&b-7\end{pmatrix} L_3 - 4L_2$ **A1**

Hence $a = 2$ and $b = 7$. **A1**

(ii) It is the same as in question (c): $r = \begin{pmatrix}2\\1\\0\end{pmatrix} + \lambda\begin{pmatrix}-2\\4\\1\end{pmatrix}$. **A1**

[5 marks]

(f) (i) $a = 2$, $b \neq 7$. **A1 A1**

(ii) $\begin{pmatrix}-2\\4\\1\end{pmatrix}\bullet\begin{pmatrix}3\\1\\2\end{pmatrix} = -6+4+2 = 0$, hence $\begin{pmatrix}-2\\4\\1\end{pmatrix} \perp \begin{pmatrix}3\\1\\2\end{pmatrix}$, hence L parallel to Π_3. **M1 A1**

[4 marks]

(a) Let R be the variable for the weight if the packet.

$\mathrm{P}(R < 480) = 0.13326... \cong 0.133$. **A1**

[1 mark]

(b) $\mathrm{P}(R < 500 \mid R > 480) = \dfrac{\mathrm{P}(480 < R < 500)}{\mathrm{P}(R > 480)} = \dfrac{0.36673...}{0.86673...} \cong 0.423$ **M1 A1**

[2 marks]

(c) $\mathrm{P}(R > a) = 0.15 \Rightarrow a = 518.65... \cong 519$ **M1 A1**

[2 marks]

(d) Let B be the variable for the number of packets under 480

Binomial with $n = 5$ and $p = 0.13326...$ **M1 A1**

$\mathrm{P}(B = 1) = 0.376$ **A1**

[3 marks]

(e) Let C be the variable for the weight of the packet of the premium variety.

(i) $Q_3 = 724 + 52 = 776$, $\mu = \dfrac{724 + 776}{2} = 750$ **M1 A1**

(ii) $\mathrm{P}(C < 724) = 0.25$. **M1**

$Z = \dfrac{C - \mu}{\sigma} \Rightarrow Z = \dfrac{724 - 750}{\sigma} \Rightarrow Z = \dfrac{-26}{\sigma}$ **M1**

From InvN(0,1): $Z = -0.67448$

$-0.67448 = \dfrac{-26}{\sigma} \Rightarrow \sigma = 38.5$ **A1**

[5 marks]

(f) $\mathrm{E}(X) = \mu = 400$, $\mathrm{Var}(X) = \sigma^2 = 25$

(i) $\mathrm{Var}(X) = \mathrm{E}(X^2) - \mu^2 \Rightarrow \mathrm{E}(X^2) = \mathrm{Var}(X) + \mu^2 = \sigma^2 + \mu^2 = 160025$ **M1 A1**

(ii) $\mathrm{E}(Y) = \mathrm{E}(3X + 2) = 3\mathrm{E}(X) + 2 = 1202$; **M1 A1**

$\mathrm{Var}(Y) = \mathrm{Var}(3X + 2) = 9\mathrm{Var}(X) = 225$ **A1**

(iii) **EITHER** as in (ii) **OR** directly since $Z = \dfrac{X - \mu}{\sigma}$ is the standardised Normal

$\mathrm{E}(Z) = 0$, $\mathrm{Var}(Z) = 1$. **A1 A1**

[7 marks]

11. [Maximum mark: 18]

(a) $(-1.20153...,1.44369...)$ so $(-1.20,1.44)$ **A1**

$(1.20153...,1.44369...)$ so $(1.20,1.44)$ **A1**

[2 marks]

(b) $\int_{-1.20153}^{1.20153...} \left(4\cos x - x^2\right)\mathrm{d}x \cong 6.30$ **M1 A1 A1**

[3 marks]

(c) $V_x = \pi \int_{-1.20153}^{1.20153...} \left(\left(4\cos x\right)^2 - x^4\right)\mathrm{d}x \qquad \left[= \pi \int_{-1.20153}^{1.20153...} \left(16\cos^2 x - x^4\right)\mathrm{d}x\right]$ **M1 A1**

$\cong 74.2$ **A1**

[3 marks]

(d) $y = 4\cos x \Rightarrow x = \cos^{-1}\frac{y}{4} \Rightarrow x^2 = \left(\cos^{-1}\frac{y}{4}\right)^2$ **A1**

$y = x^2 \Rightarrow x^2 = y$

$V_y = \pi \int_0^{1.44369} y\mathrm{d}y + \pi \int_{1.44369}^{4} \left(\cos^{-1}\frac{y}{4}\right)^2 \mathrm{d}y \cong 8.83$ **M1 A1 A1**

[4 marks]

(e) Base = $2a$, height = $h = 4\cos a - a^2$. **A1**

$A = 2a(4\cos a - a^2) = 8a\cos a - 2a^3$. **A1**

[2 marks]

(f) (i) $\frac{\mathrm{d}A}{\mathrm{d}a} = 8\cos a - 8a\sin a - 6a^2 = 0$ **M1 A1**

$8\cos a - 8a\sin a = 6a^2 \Leftrightarrow \frac{\cos a - a\sin a}{a^2} = \frac{3}{4}$

(ii) $A_{\max} \cong 3.60$. [directly from the graph of A] **M1 A1**

[4 marks]